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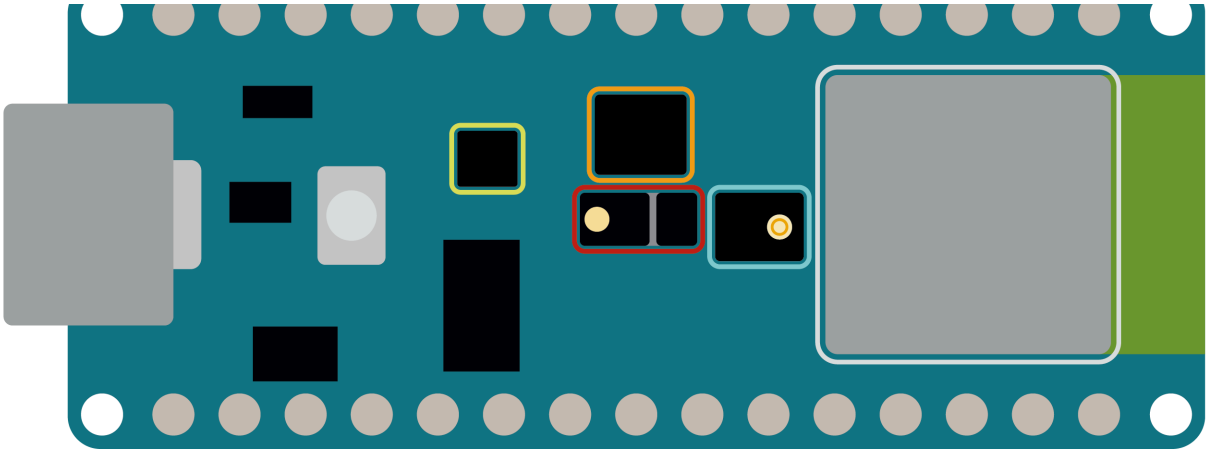
Use the built-in sensors on Nano 33 BLE Sense

The Nano 33 BLE Sense has a set of sensors embedded on the top, which gives a lot of options to try ideas without having to generate the circuit to such sensors in a prototyping board. The sensor location can be checked in the following image, while the specific model and some other details are addressed later in this same article:



- Color, brightness, proximity and gesture sensor
- Digital microphone
- Motion, vibration and orientation sensor
- Temperature, humidity and pressure sensor
- Arm Cortex-M4 microcontroller and BLE module

Color, brightness, proximity and gesture sensor

The Gesture sensor is the APDS9960 module, here you can check its datasheet. It senses gesture, color, ambience illumination and proximity. This chip, made by Broadcom, is supported by our ArduinoAPDS9960 library which includes functions and example sketches.

Digital microphone

The digital microphone is a MP34DT05, this chip, made by ST Microelectronics, is an ultra-compact, low-power, omnidirectional, digital MEMS microphone built with a capacitive sensing element and an IC interface; it produces an output coded in PDM. The PDM format is supported by our PDM library that can be used also with our ArduinoSound library which contains example sketches to use the sensor.

ST Microelectronics, is a standard component supported by our library ArduinoLSM9DS1 which contains example sketches to use the sensor. The library takes care of the sensor initialization and sets its values as follows:

- Accelerometer range is set at [-4,+4]g -/+0.122 mg
- Gyroscope range is set at [-2000, +2000] dps +/-70 mdps
- Magnetometer range is set at [-400, +400] uT +/-0.014 uT
- Accelerometer Output data rate is fixed at 104 Hz
- Gyroscope Output data rate is fixed at 104 Hz
- Magnetometer Output data rate is fixed at 20 Hz

Relative humidity and temperature

The relative humidity and temperature sensor is a HTS221, is an ultra-compact sensor that uses a polymer dielectric planar capacitor structure capable of detecting relative humidity variations and temperature, returned as digital output on a serial interface. This chip, made by ST is supported by our library ArduinoHTS221. The library contains, as usual, useful functions and example sketches to use the sensor to measure the relative humidity. The sensor can be calibrated using the function `readHTS221Calibration()`, and it's ranges are as following:

- Humidity accuracy: ± 3.5% rH, 20 to +80% rH
- Humidity range: 0 to 100 %
- Temperature accuracy: ± 0.5 °C,15 to +40 °C
- Temperature range: -40 to 120° C

Pressure

The barometer sensor is a LPS22HB, is an ultra-compact sensor which functions as a digital output barometer. This chip, made by ST is supported by our ArduinoLPS22HB library which includes example sketches to measure the atmospheric pressure.

- Absolute pressure range: 260 to 1260 hPa

If you would like to make a quick test of all the sensors, you can use this sketch to check if all the sensors are being initialized.

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